

PROJECT SCOPE AND IMPLEMENTATION

Proctoring – Student Identity Verification – Dynamic Embedding – Advanced RAG – Structured Analysis

Group Name: Project Dilemma

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1. Project Overview

This project implements a standalone, prototype-grade offline AI proctoring system that reviews recorded examination sessions (video, audio, and a browser event log) and produces a structured, evidence-backed report for a human reviewer. The system does not make a final determination of academic dishonesty. Instead, it detects observable events, verifies student identity against enrollment data, aggregates evidence into a deterministic risk tier, and — only for sessions that land in the Medium or High risk tier — uses an LLM to synthesize the evidence into a human-readable recommendation grounded in the institution's academic integrity policy via Retrieval-Augmented Generation (RAG). A human reviewer always issues the final outcome.

The project is built as a single, complete, demonstrable pipeline. Within that pipeline, the Video Detection Agent (MediaPipe + YOLOv8-nano) is implemented in the greatest depth, as per the assigned focus area; the remaining modules (audio, behavior, identity, RAG, orchestration, reporting) are implemented at a scope sufficient to make the Video Detection Agent's output meaningful and demonstrable end-to-end.

2. Problem Statement

Manual review of every proctoring recording does not scale, while fully automated cheating-detection systems are unreliable and difficult to justify ethically or academically — computer-vision and audio signals are noisy, and a false accusation carries real consequences for a student. The problem this project addresses is narrower and more defensible: how to automatically extract structured, auditable evidence from a session, verify that the person taking the exam is the enrolled student, and surface only the sessions that genuinely warrant a human's attention — with a clear, evidence-linked explanation of why — while leaving the actual judgment to a human.

3. Project Objectives

- Build a deterministic Video Detection Agent (MediaPipe + YOLOv8-nano) that converts raw exam video into timestamped, structured observations (face presence, person count, phone/object detection, gaze/pose deviation).
- Implement student identity verification using face embeddings, matching session frames against enrollment reference images with a defined confidence threshold.
- Implement a session-level dynamic embedding strategy that aggregates multiple verified-frame embeddings into a robust per-session identity representation, without permanently modifying the student's enrolled identity record.
- Implement lightweight audio and behavioral analysis agents (voice activity + basic speaker consistency; keystroke/mouse/tab-switch analytics) as supporting deterministic evidence sources.
- Implement a deterministic Risk Scoring Agent that aggregates all upstream structured evidence into a three-tier risk score (Low / Medium / High) using a transparent, explainable weighting scheme.
- Implement an Advanced RAG pipeline over academic integrity and proctoring policy documents, used by a Human Review Assistant Agent to ground its recommendation in institutional policy.
- Implement a LangGraph-based supervisor that deterministically orchestrates the agent pipeline and cost-gates LLM agents to Medium/High risk sessions only.
- Maintain structured, schema-defined records at every pipeline stage (detections, events, evidence, risk factors, final report) to support auditability and reproducibility, and present outcomes on a human review dashboard where a person issues the final decision.

4. Proposed Solution

The system is organized as a supervisor-routed multi-agent pipeline, following the provided architecture. After an exam session's video, audio, and event log are uploaded, the LangGraph Capture Orchestrator — a deterministic,

model-free router — calls four deterministic tool-agents in sequence: Video Detection, Audio Detection, Behavior Analysis, and Risk Scoring. Each agent reads and writes to a shared ProctoringState object rather than communicating directly with the others, which keeps the pipeline auditable and easy to test agent-by-agent.

If the resulting risk tier is Medium or High, the orchestrator additionally invokes two LLM-based agents: a Risk Synthesis Agent that explains the score, and a Human Review Assistant Agent that retrieves relevant policy passages via RAG and drafts a recommendation. For Cleared (Low-risk) sessions, a template-only Report Narrative Agent produces the integrity report without any LLM call, keeping per-student cost low. Every session ends on a human review dashboard, where a reviewer approves, warns, or escalates — this decision, not any agent output, is the recorded final outcome.

5. Scope of the Project

5.1 What is the project?

A standalone, offline, prototype proctoring-assistance pipeline that analyzes a recorded exam session and produces structured evidence plus a human-reviewable recommendation. It is a decision-support tool, not an autonomous decision-maker.

5.2 What problem does it solve?

It reduces the manual review burden by pre-processing every session into structured evidence and routing only ambiguous (Medium/High-risk) sessions toward LLM-assisted, policy-grounded review, while every session remains auditable.

5.3 Who is it intended for?

Academic institutions and course instructors running remote or recorded in-person examinations, and the human proctors/reviewers who make final integrity decisions.

5.4 What inputs does it take?

- A recorded exam session video file (webcam feed).
- A recorded or extracted audio track.
- A browser-side event log (tab switches, key/mouse activity timestamps).
- Enrollment reference images for each registered student, captured prior to the exam.

5.5 What does it process?

The system processes the video into per-frame/per-window detections (faces, persons, objects, pose), matches detected faces against enrollment embeddings, converts the audio into voice-activity segments, converts the event log into behavioral features, and aggregates all of this into a single structured risk record per session.

5.6 What outputs does it produce?

- A structured, machine-readable risk record per session (JSON) covering detections, identity confidence, behavioral flags, and the risk tier.
- A per-student integrity report (PDF) — template-based for Cleared sessions, LLM-narrated for Medium/High sessions.
- A reviewer-facing recommendation (Medium/High only) citing specific evidence and relevant policy passages.
- A review dashboard showing pending sessions and the final human-issued outcome (Cleared / Warning / Escalated).

5.7 Major capabilities

- Deterministic video event detection (MediaPipe + YOLOv8-nano).
- Face-embedding-based identity verification with a defined threshold and session-level (dynamic) embedding aggregation.
- Deterministic audio and behavioral signal extraction.
- Transparent, weighted risk scoring.
- Policy-grounded RAG for reviewer support.
- LangGraph-based deterministic orchestration with LLM cost-gating.
- Structured, schema-based storage for full auditability.

5.8 Technologies / concepts demonstrated

- Computer vision (object/person/face detection, pose estimation) via MediaPipe and YOLOv8-nano.
- Face recognition / embedding-based verification.
- Deterministic multi-signal risk aggregation.
- Retrieval-Augmented Generation (semantic retrieval over policy documents).
- Agentic orchestration with a state-machine supervisor (LangGraph).
- Structured data modeling for auditability.
- Prompt engineering for cost-gated, policy-grounded LLM synthesis.

5.9 Explicitly outside the scope

- The system does NOT determine, claim, or record whether a student actually cheated — it only surfaces evidence and a risk tier; the outcome is always human-issued.
- No real-time / live proctoring during the exam — processing is offline, post-session.
- No training of custom detection models from scratch — pretrained YOLOv8-nano and MediaPipe models are used as-is or lightly fine-tuned.
- No production-grade authentication, multi-tenant deployment, or scaling infrastructure.
- No gaze-tracking-based eye-movement cheating inference, lip-reading, or screen-content analysis — these require substantial custom modeling and are out of scope.
- No permanent modification of a student's enrolled identity embedding based on exam-session data.

6. Major Functional Modules

Each module below specifies its function, I/O, technology, whether it is deterministic or LLM-based, and its scope tier (MVP or Advanced).

Module	Function	Input → Output	Technology	Type
Student Enrollment	Capture and store reference face images per student	Reference images → stored embeddings	MediaPipe/InsightFace	Deterministic
Session Input Handler	Accept uploaded video/audio/event log for a session	Raw files → validated session bundle	Python, FastAPI	Deterministic
Video Detection Agent	Detect faces, persons, objects, pose per frame/window	Video → structured detections	MediaPipe + YOLOv8-nano	Deterministic
Identity Verification	Match detected faces to enrollment embeddings	Face crops → identity confidence	Face embedding model + cosine similarity	Deterministic
Dynamic Embedding Aggregator	Combine multiple verified-frame embeddings into session identity vector	Frame embeddings → session embedding	Weighted/confidence-based averaging	Deterministic

Module	Function	Input → Output	Technology	Type
Audio Detection Agent	Detect voice activity and basic speaker consistency	Audio → VAD segments, speaker-change flags	Silero VAD + Resemblyzer	Deterministic
Behavior Analysis Agent	Extract tab-switch/key/mouse activity patterns	Event log → behavioral features	Rule-based feature extraction	Deterministic
Risk Scoring Agent	Aggregate all evidence into a 3-tier risk score	Structured evidence → risk tier + score	Weighted rule-based aggregation	Deterministic
Advanced RAG Module	Retrieve relevant policy passages for a session's evidence	Query + evidence → ranked policy passages	Chroma/FAISS + embeddings	Deterministic retrieval
Risk Synthesis Agent	Explain why a session scored Medium/High	Risk record → natural-language explanation	LLM API	LLM (cost-gated)
Human Review Assistant	Draft a policy-grounded recommendation for the reviewer	Evidence + retrieved policy → draft recommendation	LLM + RAG	LLM (cost-gated)
Report Narrative Agent	Generate per-student PDF report	Risk record → PDF report	Template engine / LLM (Medium+ only)	Mixed
LangGraph Orchestrator	Route session through agents deterministically	ProctoringState → next agent call	LangGraph	Deterministic (model-free)
Review Dashboard	Display sessions, evidence, and recommendation for human decision	Session records → reviewer UI	Streamlit/React	N/A (UI)

7. Video Detection Agent — Detailed Scope

This is the assigned focus module and receives the greatest implementation depth. It is a deterministic tool agent: it detects and records observable facts from video frames and produces no judgments.

7.1 Core capabilities (in scope)

- Face detection per sampled frame (MediaPipe Face Detection / Face Mesh).
- Person detection and person count per frame (YOLOv8-nano) — flags zero-person and multiple-person frames.
- Phone / electronic-device detection using YOLOv8-nano's COCO-pretrained classes (e.g., 'cell phone').
- Basic head pose / gaze-direction estimate from MediaPipe Face Mesh landmarks, used only as a coarse 'looking away' indicator, not a precise gaze tracker.
- Student presence/absence timeline: continuous windows where no face is detected.
- Face visibility quality flag (partially occluded / poor lighting) to support downstream confidence weighting rather than hard rejection.
- Temporal event aggregation: converting per-frame detections into discrete, timestamped events (e.g., 'second person present, 04:12–04:38').
- Evidence extraction: saving representative frame crops for flagged events, referenced by ID in the structured record for reviewer inspection.

7.2 Optional / advanced capabilities (time-permitting)

- Lightweight face/person tracking across frames (e.g., IoU-based) to avoid double-counting the same event.
- Hand-detection-based heuristic for 'reaching off-screen' behavior.
- Configurable frame-sampling rate tuning for accuracy/speed trade-off.

7.3 Explicitly not implemented

- Custom-trained object classes beyond YOLOv8-nano's pretrained COCO set (no custom dataset collection for niche cheating-object detection).
- Fine-grained gaze-point estimation (screen-coordinate gaze tracking).
- Lip-reading or speech-from-video inference.

The distinction maintained throughout: object/person detection (what YOLOv8-nano and MediaPipe do) is separate from face recognition/identity verification (a distinct embedding-matching step, described in Section 8). The Video Detection Agent never claims to know who a detected face belongs to — it only reports that a face was detected, with what confidence and quality.

8. Student Identity Verification — Scope

- Enrollment: each student uploads 2–3 reference images once, prior to any exam session.
- Reference embedding generation: face is detected and cropped (MediaPipe), then passed through a dedicated face-embedding model (recommended: InsightFace/ArcFace via the insightface Python package) to produce a stored reference vector per student.
- Embedding storage: reference embeddings stored in a lightweight vector store keyed by student ID (see Section 10).
- Face matching: during session processing, each verified face crop from the Video Detection Agent is embedded and compared to the student's stored reference via cosine similarity.
- Verification threshold: a fixed similarity threshold (tuned empirically, e.g., ~0.5–0.6 depending on the embedding model) classifies a frame as matched / uncertain / mismatched.
- Handling unknown/multiple faces: any face that does not match the enrolled student above threshold is logged as an 'unverified face' event, distinct from a 'second person present' detection event, and both are surfaces to the reviewer separately.
- Session-level identity confidence: an aggregate confidence score (e.g., percentage of sampled frames with a verified match) rather than a single frame-level pass/fail, since lighting and pose vary naturally through a session.

This module is intentionally kept as embedding + threshold + aggregation — no custom face-recognition model training is required, which keeps it realistic for a single-student timeline.

9. Dynamic Embedding — Scope Definition

“Dynamic Embedding” in this project means session-level, temporary embedding aggregation — not continuous modification of a student's permanent enrolled identity.

- Multiple face embeddings are extracted from verified, high-quality frames throughout a session (not just once at the start).
- These per-frame embeddings are aggregated (confidence-weighted average, discarding low-quality or low-confidence frames) into a single session-representative embedding.
- This session embedding is used only to compute the session's overall identity-confidence score; it is discarded after the session is scored — it is a temporary, session-scoped representation, not a persistent update.
- Identity drift prevention: the student's enrollment reference embedding is never overwritten by session data. Any proposal to update enrollment data requires a separate, explicit re-enrollment action, never an automatic one triggered by exam-session processing.

This gives a technically defensible, implementable meaning to 'dynamic' — the embedding used for matching is dynamically constructed per session from multiple observations (improving robustness to a single bad frame), while the enrolled identity itself stays fixed and auditable.

10. Advanced RAG — Scope

10.1 Knowledge sources

- Academic integrity policy documents (institution-provided or representative sample text).
- Proctoring guidelines (allowed/disallowed items, permitted breaks, etc.).
- Reviewer guidelines / escalation criteria.

10.2 What is retrieved, and why

Given a session's structured risk record (e.g., 'second person detected for 26 seconds, phone detected once, identity confidence 71%'), the Human Review Assistant Agent queries the policy knowledge base for passages relevant to those specific evidence types, so its draft recommendation cites the applicable policy rather than a generic summary.

10.3 Recommended retrieval techniques (right-sized for a student project)

- Semantic retrieval via sentence-embedding similarity search (in scope, MVP).
- Metadata filtering, e.g., retrieving only 'device policy' vs 'presence policy' sections based on which evidence types fired (in scope, MVP — this is what earns the 'Advanced' label without heavy infrastructure).
- Query construction from structured evidence rather than free text, i.e., the query is built programmatically from the risk record's fields (in scope, MVP).
- Reranking and query rewriting (optional, time-permitting — not essential to demonstrate the concept).
- Hybrid (keyword + semantic) retrieval (optional, only if the policy corpus is large enough to need it).

Deliberately not pursued: a full production RAG stack (managed vector DB, reranker service, evaluation harness). A local FAISS or Chroma index with metadata filtering is sufficient to demonstrate 'advanced' retrieval logic without over-engineering.

11. Structured Analysis — Scope

The system maintains explicit, schema-defined records at every stage rather than free-text conclusions, so every claim in a report traces back to a specific structured fact.

Record type	Purpose	Key fields
Detection	Single per-frame CV output	frame_ts, type, bbox, confidence
Observation	Aggregated detection over a time window	start_ts, end_ts, type, count
Event	Discrete flaggable occurrence	event_id, type, start_ts, end_ts, evidence_ref
Evidence	Supporting artifact for an event	evidence_id, event_id, frame_path/audio_clip
Behavioral Pattern	Aggregated behavior-log feature	pattern_type, frequency, session_id
Risk Factor	Single weighted contributor to score	factor_type, weight, source_event_ids
Risk Score	Aggregated session outcome	session_id, tier, numeric_score, factor_breakdown
Final Report	Reviewer/student-facing document	session_id, narrative, cited_events, cited_policy

Record type	Purpose	Key fields
Review Outcome	Human decision	session_id, reviewer_id, decision, timestamp, notes

Every downstream artifact (risk score, LLM narrative, reviewer dashboard entry) references these records by ID rather than restating facts in prose, which is what makes the system auditable: a reviewer (or grader) can trace any claim in a final report back to the exact frame, timestamp, or policy passage that produced it.

12. Agentic Architecture

The Capture Orchestrator is implemented as a LangGraph graph over a shared ProctoringState. It is model-free: it reads state fields written by the previous agent and follows conditional edges — Video → Audio → Behavior → Risk Scoring runs for every session; Risk Synthesis and Human Review Assistant fire only when Risk Scoring's output tier is Medium or High; Report Narrative branches between its template path (Cleared) and LLM path (Medium/High).

This cost-gating is a core design decision, not an afterthought: it keeps the LLM-dependent portion of the pipeline proportional to genuinely ambiguous sessions, which is both economically realistic and a better academic story (“we used generative AI where it adds value, and deterministic tools where they suffice”) than routing every session through an LLM.

13. Technology Stack

Layer	Technology	Used by	Essential?	Tier
Language	Python 3.11	All backend modules	Yes	MVP
Computer Vision	OpenCV	Frame extraction/preprocessing	Yes	MVP
Object Detection	YOLOv8-nano (Ultralytics)	Video Detection Agent	Yes	MVP
Face/Pose Landmarks	MediaPipe	Video Detection Agent, Identity Verification	Yes	MVP
Face Embedding	InsightFace (ArcFace)	Identity Verification, Dynamic Embedding	Yes	MVP
Audio VAD	Silero VAD	Audio Detection Agent	No	Advanced
Speaker Embedding	Resemblyzer	Audio Detection Agent	No	Advanced
Agent Orchestration	LangGraph	Capture Orchestrator	Yes	MVP
Backend API	FastAPI	Session upload, agent invocation endpoints	Yes	MVP
Relational Storage	SQLite (dev) / PostgreSQL (optional)	Structured records, enrollment data	Yes	MVP
Vector Storage	Chroma or FAISS	Face embeddings, policy document embeddings	Yes	MVP
LLM	Anthropic Claude API (or equivalent)	Risk Synthesis, Human Review Assistant, Report Narrative	Yes	MVP
Frontend / Dashboard	Streamlit	Review dashboard	Yes	MVP
Reporting	ReportLab / WeasyPrint	PDF report generation	Yes	MVP

Rationale for the simplest viable stack: Streamlit over React avoids building a separate frontend service while still supporting an interactive reviewer dashboard; SQLite avoids standing up a database server for a single-developer prototype while remaining a drop-in swap for PostgreSQL later; Chroma/FAISS are chosen over a managed vector DB because the embedding volumes in a student project (a few hundred sessions, a small policy corpus) don't need managed infrastructure.

14. Implementation Methodology

Phases are ordered so that each phase produces a testable artifact the next phase depends on. Video Detection is prioritized early (Phase 3) since it is the assigned focus and the primary evidence source everything else builds on.

Phase	Objective	Key tasks	Output	Depends on
1. Setup	Establish project skeleton and data schema	Repo structure, ProctoringState schema, SQLite schema, sample video acquisition	Runnable skeleton + schema	—
2. Video pipeline	Frame extraction and preprocessing	OpenCV frame sampling, resizing, storage of frame refs	Extracted frame set per session	Phase 1
3. Video Detection Agent	Core assigned module	YOLOv8-nano integration, MediaPipe face/pose, event aggregation logic	Structured detections + events JSON	Phase 2
4. Identity verification	Face matching against enrollment	InsightFace embedding, enrollment flow, threshold tuning	Per-frame + session identity confidence	Phase 3
5. Dynamic embedding	Session-level embedding aggregation	Confidence-weighted aggregation logic	Session embedding + confidence score	Phase 4
6. Audio + behavior (Advanced)	Supporting deterministic signals	Silero VAD, Resemblyzer, event-log feature extraction	Audio/behavior structured records	Phase 1
7. Risk scoring	Deterministic aggregation	Define weights, implement scoring function, tier thresholds	Risk Score record	Phases 3–6
8. RAG module	Policy retrieval	Chunk/embed policy docs, build Chroma/FAISS index, metadata filters	Retrieval function	Phase 1
9. LLM agents	Synthesis + review assistant	Prompt design, cost-gating logic, RAG-grounded generation	Narrative + recommendation	Phases 7–8
10. LangGraph orchestration	Wire all agents into supervisor graph	Define graph nodes/edges, conditional routing, state passing	End-to-end pipeline	Phases 3–9
11. Report + dashboard	Reviewer-facing outputs	PDF generation, Streamlit dashboard, review-decision recording	Working UI + reports	Phase 10
12. Testing & evaluation	Validate against sample scenarios	Curated test videos, metric collection, threshold tuning	Evaluation results	Phase 11

15. MVP, Advanced, and Future Scope

15.1 MVP / Core Scope (must implement)

- Video Detection Agent: face detection, person count, phone/object detection, presence timeline (MediaPipe + YOLOv8-nano).
- Identity verification with enrollment, embedding, threshold-based matching, and session-level confidence.
- Dynamic (session-level) embedding aggregation.
- Deterministic Risk Scoring Agent with transparent weighting.
- RAG over a small policy corpus with semantic retrieval + metadata filtering.
- Human Review Assistant Agent (LLM + RAG) for Medium/High sessions.
- Template-based report for Cleared sessions; LLM-narrated report for Medium/High.
- LangGraph orchestrator with cost-gated conditional routing.
- Structured records for detections, events, risk factors, and final reports.
- Streamlit review dashboard with human-issued final outcome.

15.2 Advanced Scope (implement if time permits)

- Audio Detection Agent (Silero VAD + Resemblyzer).
- Behavior Analysis Agent (tab/key/mouse analytics).
- Risk Synthesis Agent (separate from Human Review Assistant) explaining the score itself.
- Lightweight face/person tracking across frames.
- Reranking or query rewriting in the RAG pipeline.

15.3 Future Scope (explicitly not promised in this submission)

- Real-time / live proctoring during the exam.
- Custom-trained detection models for niche cheating-related objects.
- Fine-grained gaze-point / screen-coordinate tracking.
- Production deployment: multi-tenant auth, horizontal scaling, managed vector DB/reranker services.
- Automated policy-corpus ingestion pipeline for arbitrary institutions.

16. Project Deliverables

- Working end-to-end proctoring pipeline (backend + dashboard) runnable on sample data.
- Video Detection Agent module with test coverage on sample videos.
- Identity verification and dynamic embedding module.
- Deterministic Risk Scoring Agent.
- RAG module with a small sample policy corpus.
- LLM-based Human Review Assistant and Report Narrative agents.
- LangGraph orchestration graph.
- Sample structured reports (PDF) for at least one Cleared and one Medium/High session.
- Streamlit review dashboard.
- Architecture diagram (updated to reflect actual implementation).
- This scope document and a brief evaluation/test report.

17. Evaluation Strategy

Component	Metrics	Test approach
Identity verification	True/false accept rate at chosen threshold	Curated set of matching + non-matching face pairs
Object/person detection	Precision/recall on sample frames (manually labeled)	Small hand-labeled evaluation set from sample videos

Component	Metrics	Test approach
Video event detection	Event boundary accuracy vs. manual annotation	Manually annotate 3–5 sample sessions, compare timestamps
Behavioral analysis	Flag rate vs. scripted behavior	Scripted test sessions with known tab-switch counts
RAG retrieval	Relevance of top-k retrieved passages (manual judgment)	Query set built from known evidence → check retrieved policy relevance
LLM output	Groundedness (no unsupported claims) via manual review	Manually check narratives cite only retrieved/structured evidence
End-to-end	Risk tier agreement with manually assessed 'expected' tier	Set of scripted test sessions with known intended outcome (clean, borderline, clear-violation)

Practical test scenarios: (1) a clean session with the enrolled student alone, well-lit, no device; (2) a session with a brief second-person appearance; (3) a session with a visible phone; (4) a session where an unenrolled person takes the exam; (5) a session with poor lighting/partial occlusion, to test graceful degradation rather than false flags.

18. Limitations

- The system does not, and does not claim to, definitively determine whether cheating occurred — all outcomes require human review.
- Processing is offline/post-session, not real-time.
- Detection models are pretrained (YOLOv8-nano, MediaPipe) and not fine-tuned on institution-specific data; accuracy is prototype-level.
- Evaluation is limited to a small, manually curated set of sample/scripted videos, not a large labeled benchmark.
- Performance is sensitive to lighting, camera angle, and video quality; low-quality input can produce false positives or false negatives.
- Object detection is limited to YOLOv8-nano's pretrained COCO classes; niche cheating aids outside this class set will not be detected.
- This is a prototype/demonstration system, not a production-hardened, security-audited deployment.

19. Expected Outcomes

A demonstrable, end-to-end prototype that ingests a recorded exam session and produces a structured, auditable risk assessment together with a human-reviewable, policy-grounded recommendation for ambiguous cases — showcasing computer vision, face-embedding-based identity verification, deterministic multi-signal risk aggregation, Retrieval-Augmented Generation, and LangGraph-based agentic orchestration within a scope one student can realistically build and defend in a viva.